

Germano Pansini

📍 Bari, Italy

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Professional Profiles, Projects, Media, and Publications are available via the Portfolio.

SKILLS

Programming : Python, C++, MATLAB

Perception / ML : PyTorch, TensorFlow, Ultralytics YOLO, SegFormer

Robotics : ROS2/ROS, PX4, SLAM, Visual Odometry, Sensor Fusion, LiDAR, RGB-D, IMU, Sonar

Simulation & Digital Twins : NVIDIA Omniverse, Isaac Sim, OpenUSD, Gazebo, RViz

CV / Data Libraries : OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib

Systems & Tooling : Linux (Ubuntu), Bash, Git, Docker, Conda, SSH, Jetson, Raspberry Pi, Pixhawk

EXPERIENCE

Post-Graduate Research Fellow - TRAPScore

Apr. 2024 – Oct. 2024

Polytechnic University of Bari

- Developed an end-to-end pest object-detection pipeline for in-field monitoring, covering dataset curation and versioning, YOLOv8 training and validation, insect counting, and spatio-temporal analysis on real agricultural data.
- Designed and deployed RGB edge platforms based on Raspberry Pi 4 for real-time inference with YOLOv8, integrating OpenCV pre/post-processing, Roboflow-driven augmentation workflows, and a Flutter dashboard for live monitoring and reporting.

Post-Graduate Research Fellow - PON AGREED

Dec. 2022 – Mar. 2023

Polytechnic University of Bari

- Brought a tracked rover from a mechanics-only prototype to a fully operational robotic system, designing power architecture, LiPo charging, motor control (Sabertooth 2x60), onboard computing, and LiDAR/RGB-D integration.
- Implemented a ROS-based autonomy stack including LiDAR SLAM (RViz/Gazebo), sonar-based obstacle detection, multi-sensor fusion, and secure remote teleoperation with joystick input and RGB streaming over Wi-Fi/SSH.

EDUCATION

M.Sc. in Computer Science Engineering (AI & Data Science)

Sep. 2022 – Dec. 2025

Polytechnic University of Bari

- **Final grade:** 106/110
- **Thesis:** *Neuro-Symbolic Semantic Segmentation Framework for Autonomous Space Exploration*

B.Sc. in Electronic and Telecommunications Engineering

Sep. 2016 – Apr. 2022

Polytechnic University of Bari

- **Final grade:** 104/110
- **Thesis:** *Development of an Integrated Hardware Platform for Autonomous Robot Navigation*

PUBLICATIONS

Crop Monitoring via Deep-Learning-Based Pest Classification for Pesticide Reduction

Nov. 2024

IEEE International Humanitarian Technologies Conference (IHTC)

- Proposed and validated a real-time YOLOv8-based multi-class pest object-detection framework on the IP102 dataset, explicitly addressing intra-class variability and pest life-cycle appearance changes for scalable crop monitoring.
- Designed dataset balancing and interleaved training strategies that improved precision and F1 scores over standard YOLO baselines, with experimental results published at IEEE IHTC.

AWARDS

Winner & Research Presenter

Oct. 2024

Maker Faire 2024 The European Edition, Rome, Italy

- Vision-driven drone control system enabling real-time gesture-based flight, integrating on-board hand-tracking perception, low-latency vision pipelines, and ROS-based control with PID stabilization for smooth and robust maneuvering.
- Tracked rover platform for remote navigation and real-time 3D mapping, integrating LiDAR-based SLAM within ROS, live dense point cloud generation, multi-sensor fusion, and reliable remote teleoperation.